

A188 · Expression of Cardiac Vagal Sensory Neuron Markers in Human Dilated Cardiomyopathy: A Reanalysis of Public scRNA-seq/snRNA-seq Data

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Vagal sensory neurons (VSNs) innervating the heart play a central role in the heart-brain axis, displaying complex molecular signatures tied to where their nerve endings land. Previous mouse studies pinpointed *Piezo2*, *Drd2*, and *Agtr1a* as key markers for these neural connections. To test whether these candidate genes alter during pathology, a re-analysis of a public human cardiac transcriptomic atlas (scRNA-seq/snRNA-seq) from 27 healthy donors and 18 patients with dilated cardiomyopathy (DCM) was performed. While *Drd2* could not be analyzed due to ID matching issues across files, *Agtr1a* was heavily expressed in pericytes and fibroblasts, dropping significantly in DCM hearts ($p_{adj} < 0.001$). *Piezo2* showed low, scattered expression across most cell types, but revealed a focal neuronal signal exclusive to DCM, including a small cell population. These results suggest potential remodeling in the perivascular space linked to vagal-cardiac signaling during heart disease, providing preliminary, hypothesis-generating evidence on the role of cardiac VSN markers in DCM.